Bohui Zhang

Contact Information

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Research Interests

My research interests lie at the intersection of knowledge graphs, natural language processing, and explainable AI, particularly mechanistic interpretability and knowledge representation in large language models. My work focuses on automated and explainable knowledge graph construction, NLP-driven and multi-agent knowledge engineering, and understanding how factual and ontological knowledge is represented and processed within LLMs.

Academic Background

King's College London (KCL) 2022 - 2026
Ph.D. in Computer Science (Thesis Submitted, Viva Pending)
Supervisors: Prof. Elena Simperl, Dr. Albert Meroño Peñuela

University of Southern California (USC) 2020 - 2021
M.S. in Applied Data Science

University of Waterloo (UWaterloo) 2015 - 2019
B.S. in Materials and Nanosciences, *Dean's Honours List*

Beijing Jiaotong University (BJTU) 2015 - 2019
B.Eng. in Nanomaterials and Nanotechnology

Work Experience

USC Information Sciences Institute, Student Researcher 09/2021 - 12/2021

- Supervisor: Dr. Filip Ilievski
- Investigated the feasibility of enriching Wikidata with structured data sources from the linked open data (LOD) cloud.
- Proposed a method that consists of several steps: gap detection, external graph selection, schema alignment, knowledge retrieval, and validation, implemented the procedure using the Knowledge Graph Toolkit (KGTK).
- Evaluated the method on enriching Wikidata with two LOD sources: DBpedia and Getty Vocabularies. The experiments showed that the LOD-based method can enrich Wikidata with millions of new high-quality statements in a short time.

Alibaba Cloud, Machine Learning Engineer Intern 05/2021 - 08/2021

- Mentor: Jingjun (Alvin) Chu
- Worked on a Neural Architecture Search (NAS) system for optimizing models in search space defined by ProxylessNAS, used on image classification and feature extraction tasks based on dataset collecting from group's retail sector.
- Improved the model training process using knowledge distillation and improved the optimal model architecture searching process in various hardware environments using policy gradient algorithm based on target accuracy, FLOPs and latency.

- The optimal models deployed on terminal machines achieved model compression for more than 60% decrease on FLOPs while improving rank1 and rank6 compared with state-of-the-art MobileNetV2 models and keeping the top1 accuracy above 98%.

Publications [8]

9. Jan-Christoph Kalo, Simon Razniewski, **Bohui Zhang**, Tuan-Phong Nguyen, Summary of the AKBC 2026 Shared Task: Language Models for Knowledge Base Construction (LM-KBC, 5th Edition), In *Workshop on Automated Knowledge Base Construction*, 2026
8. **Bohui Zhang**, Yuan He, Lydia Pintscher, Albert Meroño-Peñuela, Elena Simperl, Schema Generation for Large Knowledge Graphs Using Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025
7. **Bohui Zhang**, Elisavet Koutsiana, Yihang Zhao, Albert Meroño-Peñuela, and Elena Simperl, Trustworthy Knowledge Graphs: Practices and Approaches. In *Handbook on Neurosymbolic AI and Knowledge Graphs*. IOS Press, 2025
6. Elisavet Koutsiana, Johanna Walker, Michelle Nwachukwu, **Bohui Zhang**, Albert Meroño-Peñuela, Elena Simperl, Knowledge Prompting: How Knowledge Engineers Use Generative AI, In *Journal of Web Semantics*, 2025
5. **Bohui Zhang**, Albert Meroño-Peñuela, and Elena Simperl, Towards Explainable Automated Knowledge Engineering with Human-in-the-loop, In *Semantic Web Journal*, 2025
4. **Bohui Zhang**, Valentina Anita Carriero, Katrin Schreiberhuber, Stefani Tsaneva, Lucía Sánchez González, Jongmo Kim, Jacopo de Berardinis, OntoChat: a Framework for Conversational Ontology Engineering using Language Models, In *Extended Semantic Web Conference*, 2024
3. **Bohui Zhang**, Ioannis Reklos, Nitisha Jain, Albert Meroño Peñuela, Elena Simperl, Using Large Language Models for Knowledge Engineering (LLMKE): A Case Study on Wikidata, In *Knowledge Base Construction from Pre-trained Language Models Workshop at International Semantic Web Conference*, 2023
2. **Bohui Zhang**, Albert Meroño Peñuela, Elena Simperl, Towards Explainable Automatic Knowledge Graph Construction with Human-in-the-loop, In *International Conference on Hybrid Human-Artificial Intelligence* (HHAI), 2023
1. **Bohui Zhang**, Filip Ilievski, Pedro Szekely, Enriching Wikidata with Linked Open Data, In *Wikidata Workshop co-located with International Semantic Web Conference*, 2022

Preprints

2. **Bohui Zhang**, Albert Meroño-Peñuela, Elena Simperl, Can LLMs Faithfully Solve Ontology Engineering Tasks? A Mechanistic Interpretability Study, *arXiv preprint arXiv:2610.38326*, 2026
1. Célian Ringwald, **Bohui Zhang**, Catherine Faron, Franck Michel, Hanna Abi Akl, Albert Meroño-Peñuela, Fabien L. Gandon, Elena Simperl, Cross-Syntax Schema Generation for Knowledge Graphs: A Paired Benchmark and Agentic Workflow, *Manuscript Under Review*, 2026

Teaching Assistant

5CCSAKNE Knowledge Engineering KCL

- Module leader: Dr. Albert Meroño Peñuela, Prof. Elena Simperl
- Developed course materials, including lecture notes and lab sheets, on knowledge graphs, knowledge engineering, and large language models; led lab sessions.
- Semesters: 2025-26 Semester 2, 2024-25 Semester 2

7CUSMNDA Network Data Analysis KCL

- Module leader: Dr. Albert Meroño Peñuela
- Designed and delivered coding lab sessions, topics covered included graph theory, spatial and social network analysis, graph embedding, and semantic web.
- Semesters: 2025-26 Semester 2, 2024-25 Semester 2, 2023-24 Semester 2, 2022-23 Semester 2

5CCS2FC2 Foundations of Computing II KCL

- Module leader: Dr. Christopher Hampson
- Delivered lab sessions on algorithm problems, topics covered included P/NP, SAT solving, approximation, linear programming, and probabilistic algorithms.
- Semesters: 2022-23 Semester 1

Awards

NMES Enterprise & Engagement Partnerships Fund	KCL, 2023
Winner (Open Track), LM-KBC Challenge @ ISWC 2023	KCL, 2023
Graduation Dean's Honours List	UWaterloo, 2019
Waterloo-Beijing Jiaotong University Tuition Award	BJTU, 2016, 2017, 2018
Excellence Scholarship of Academic Activities	BJTU, 2017 - 2018
Excellence Scholarship of Social Activities	BJTU, 2015 - 2016

Academic Service

Organizer for ELMKE Workshop series, LM-KBC Challenge series, Knowledge Prompting Hackathon 2023.
Editorial Board Member for Journal of Web Semantics.
Reviewer for Information Processing & Management, Semantic Web, Information, AAAI 2026, ISWC 2024, ACM CHI 2024, 2023.
Member of Knowledge Graphs Interest Group at the Alan Turing Institute.

Skills

Programming Languages: Python, Java, JavaScript, MATLAB
Frameworks & Platforms: Docker, LangChain, PyTorch, transformers, KGTK, LangGraph
Semantic Web Tech Stacks: OWL, RDF, SHACL, ShEx, SPARQL, PROV
Databases: MongoDB, MySQL, Neo4j